How becoming ESG rated affects firm’s leverage ratios and debt structure

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Introduction

- ESG considerations have received growing attention
- Previous studies access the impact of ESG on several determinants of firms’ capital structure but with limitations
- Filling the gap and studying the of ESG on firms’ debt structure
- How becoming ESG rated affects firm’s leverage ratios and debt structure
- Assessing firms’ overall leverage ratios and debt elements (i.e. bank versus bond borrowing)

Previous Studies

- Firms with superior ESG performance face lower capital constraints (Cheng et al., 2014)
- Social strengths may reduce cost of equity (El Ghoul et al., 2011; Ng and Rezaee, 2015)
- Environmental strengths will not affect on the cost of equity (Chava, 2014)
- Firms invest in positive ESG activities are charged in all dimensions pay about 25 bps higher (Goss and Roberts, 2011)
- Cost of bonds for socially responsible firms are higher (Menz, 2010)
- Environmental risk management increases the leverage ratios (Sharfman and Fernando, 2018)
- Fair employee treatment or higher employee well-being decreases debt ratios (Verwijmeren and Derwall, 2010)
- ESG performance influences firms' speed of adjustment towards target leverage ratios (Ho et al., 2021)

However:

The level of leverage?

How becoming ESG rated affects firms' debt structure directly?

Previous Studies Continue…

- Capital structure is irrelevant to firm value under a perfect capital market (Modigliani and Miller, 1958)
- Trade-off theory: Firms take advantage of tax shiels from debt financing (Kraus and litzenberger, 1973)
- It is possible to narrow the gap between actual leverage ratio and the target leverage ratio (Ju et al., 2015)
- Pecking order theory: Asymmetric information issues. Firms borrow starts from internal cash, then debt and at the end equity (Myers and Majluf, 1984)
- Firms seek more internal safer sources due to asymmetric information (Denis and Mihov, 2003)
- Bank loans are regarded as inside debt (James, 1987)
- Debt overhang could lead to underinvestment issues (Myers, 1977)

However:

How these theories matter in debt (re-)structuring and financing decisions when firms become ESG rated?

Hypothesis

- Firms with (high) ESG ratings have higher growth opportunities (Lins et al., 2017)
- Stakeholders face high switching cost if firms are liquidated (Titman, 1984)
- ESG-rated firms are facing lower costs of bank loans and bonds (El Ghoul et al., 2011)

Hypothesis I: Firms that become ESG rated exhibit lower target leverage ratios

- Compared to issuing bonds, bank loans are treated as inside debt (James, 1987)
- Borrowing from banks exhibits less asymmetric information (Besanko and Kanatas, 1993)
- Borrowers who use bank loans can avoid floatation costs of bond issues (Easterwood and Kadapakkam, 1991)
- Lenders (banks) are eager to maintain a long-haul business relationship with ESG-rated firms (Bacha and Ajina, 2019)
- To attract good firms, banks might lower collateral and covenant requirements (Hasan et al., 2017)

Hypothesis II: Firms that become ESG rated are able to redistribute financing from external to internal sources, i.e. from issuing bonds to bank loans

Data and Processing

Data:
Refinitiv: Firm-level ESG ratings
Capital IQ: Debt structure information
CRSP-Compustat merged (CCM) annual database: firm-level data to describe firm characteristics

Sample:

Table 1
The number of remaining firms and firm-year observations.

Process	Firm			Firm-year observations		
	With	Without	Total	With	Without	Total
After merging	437	4,425	4,762	2,238	34,683	36,921
Drop, non-common firms	304	3,262	3,566	1,590	24,402	25,992
Drop, if assets=. or =0	304	3,262	3,540	1,588	23,422	25,010
Drop, if debts=. or =0	292	3,153	3,445	1,520	21,880	23,400
Drop, if bl(ml) < 0 or > 1	290	2,894	2,923	1,491	16,500	17,991
Drop, if year < 2002	290	2,550	2,840	1,491	15,046	16,536
Drop, if difference > 10%	283	2,454	2,737	1,385	13,432	14,817
Drop, if debt ratios > 1	250	,2097	2,347	1,132	9,886	11,018
Winsor 1% and 99%	250	2,097	2,347	1,132	9,886	11,018

Notes: This table shows the process of data clean. The first column describes conditions used to drop observations. Here, we drop observations in financial firms and utilities, without assets and debts or values of assets and debts are missing, are outside of unit interval, and are before the year 2002. Also, observations with total debt in CCM and Capital IQ databases exceeds 10% or any debt ratio exceeds one are deleted. We winsorize top and bottom 1% values. The second column to the fourth column introduce the number of rated firms, non-rated firms, and total firms respectively. The fifth to the seventh columns show the number of remaining firm-year observations with ESGC scores, without ESGC scores and the number of all observations for each. In the end, there are 11,018 firm-year observations with 2,347 unique firms from 2002 to 2019.

Variables description

Table 2
Variables description.

Variables	Description	Source
Debt structure		
Market leverage	Long-term debt (9)/((assets (6) − common equity (60)) + price close (31) ∗ common shares outstanding (25))	
Book leverage	Long-term debt (9)/assets (6)	CCM
Bank debt ratio	(Revolving credit + term loans)/debt	
Bond debt ratio	(Senior bonds notes + subordinated bonds notes)/debt	
Commercial paper ratio	Commercial paper/debt, where debt = debt in current liabilities (34) + long-term debt (9)	
Revolving credit ratio	Revolving credit/debt	
Term loans ratio	Term loans/debt	
Senior bonds & notes ratio	Senior bonds & notes/debt	Capital IQ & CCM
Subordinated bonds & notes ratio	Subordinated bonds & notes/debt	
Capital leases ratio	Capital leases/debt	
Other debt ratio	(other debt + total trust-preferred stock)/debt	
ESG indicators		
Ln(ESGC)	Natural logarithm of ESG combined score	
Ln(ESG)	Natural logarithm of ESG score	
Ln(EP)	Natural logarithm of environment pillar score	Refinitiv ESG
Ln(SP)	Natural logarithm of social pillar score	
Ln(GP)	Natural logarithm of governance pillar score	
Firm characteristics		
Assets	Natural logarithm of assets (6)	
Market-to-Book ratio	Market value of equity divided by book value of equity, i.e. price close (31) ∗ common shares outstanding (25)/(stockholders' equity (144) + deferred taxes (74) + investment tax credit (208) -preferred_stock), where preferred_stock = pstkrv (56) (if missing, use pstkl (10); if still missing, use pstk (130))	
Sales	Natural logarithm of sales (12)	
Tangibility	Property, plant, and equipment (8)/assets (6)	CCM
Profitability	Operating income before depreciation (13)/assets (6)	
R&D expense	Research and development expense (46)/sales (12)	
SGA cost	Selling, general and administrative expenses (132)/sales (12)	
Dividend	Dummy = 1, if dividend payment (21) = 0	
Sales over assets	Sales (12)/assets (6)	
Financial pressure	Cash flow divided by interest payments, i.e. (operating income before depreciation (13) − interest and related expense (15) − income taxes (16))/interest and related expense	
R&D intensity	R&D expenses (46)/number of employees (29)	

Descriptive Analyzing

Table 3
Descriptive statistics (full sample).

	N	Mean	P25	P50	P75	Std. Dev.
Debt structure variables						
Market leverage	10,981	0.151	0.005	0.090	0.243	0.172
Book leverage	11,018	0.212	0.012	0.153	0.350	0.216
Bank/debt	11,018	0.535	0.044	0.586	0.984	0.410
Bond/debt	11,018	0.335	0	0.118	0.707	0.385
Commercial paper/debt	11,018	0.002	0	0	0	0.030
Revolving credit/debt	11,018	0.217	0	0	0.336	0.338
Term loans/debt	11,018	0.319	0	0.083	0.655	0.384
Senior b&n/debt	11,018	0.286	0	0.006	0.603	0.375
Subordinated b&n/debt	11,018	0.049	0	0	0	0.170
Capital leases/debt	11,018	0.098	0	0	0.021	0.260
Other debt/debt	11,018	0.040	0	0	0	0.161
ESG variables						
Ln(ESGC)	11,018	0.348	0	0	0	1.043
Control variables						
Ln(asset)	11,018	5.418	3.912	5.553	7.083	2.356
Market-to-Book ratio	10,681	2.523	0.824	1.732	3.425	11.402
Ln(sale)	10,455	5.054	3.638	5.270	6.731	2.457
Tangibility	11,014	0.315	0.063	0.188	0.539	0.299
Profitability	10,998	−0.069	−0.056	0.077	0.136	0.504
R&D expense	11,018	0.372	0	0	0.051	1.889
SGA cost	9,550	0.648	0.101	0.225	0.473	1.929
Dividend	11,018	0.232	0	0	0	0.422
Sales/assets	11,018	0.876	0.292	0.657	1.203	0.829

Notes: This table provides descriptive statistics of the main variables under the full sample. Data comes from merged CRSP-Compustat, Capital IQ, and Refinitiv databases between 2002 and 2019. Variables are split into debt structure variables, ESG variables, and control variables. N denotes the number of firm-year observations. Noting that senior b&n and subordinated b&n means senior and subordinated bonds and notes; Research and Development expenditure is called R&D expense; SGA cost is Selling, General, and Administrative expense.

Descriptive Analyzing Continue...

Table 4
Descriptive statistics (matched sample).

	N	Mean	P25	P50	P75	Std. Dev.
Debt structure variables						
Market leverage	4,322	0.169	0.025	0.130	0.259	0.163
Book leverage	4,328	0.243	0.049	0.212	0.379	0.211
Bank/debt	4,328	0.521	0.068	0.540	0.971	0.401
Bond/debt	4,328	0.361	0	0.228	0.728	0.383
Commercial paper/debt	4,328	0.003	0	0	0	0.031
Revolving credit/debt	4,328	0.198	0	0	0.272	0.319
Term loans/debt	4,328	0.323	0	0.118	0.637	0.377
Senior b&n/debt	4,328	0.309	0	0.046	0.647	0.376
Subordinated b&n/debt	4,328	0.052	0	0	0	0.175
Capital leases/debt	4,328	0.090	0	0	0.018	0.250
Other debt/debt	4,328	0.036	0	0	0	0.151
ESG variables						
Ln(ESGC)	4,328	0.870	0	0	2.262	1.506
Control variables						
Ln(asset)	4,328	6.499	5.252	6.546	7.855	1.966
Market-to-Book ratio	4,144	2.756	1.109	2.028	3.571	7.589
Ln(sale)	4,300	6.116	4.880	6.288	7.518	2.085
Tangibility	4,328	0.270	0.064	0.167	0.416	0.260
Profitability	4,328	0.089	0.061	0.109	0.156	0.155
R&D expense	4,328	0.206	0	0	0.046	1.366
SGA cost	3,989	0.318	0.096	0.204	0.365	0.849
Dividend	4,328	0.292	0	0	1	0.455
Sales/assets	4,328	0.903	0.436	0.755	1.211	0.661

Notes: This table provides descriptive statistics for the matched sample. Data is from merged CRSP-Compustat, Capital IQ, and Refinitiv databases between 2002 and 2019. The variables are split into debt structure variables, ESG variables, and control variables. N denotes the number of firm-year observations. Noting that senior b&n and subordinated b&n means senior and subordinated bonds and notes; MB ratio is Market-to-Book ratio; SGA cost is Selling, General, and Administrative expense.

Empirical Analysis

$$d_{i,t}^{*} = \alpha + B'\mathbf{X}_{i,t} + \eta_i$$

$$d_{i,t} - d_{i,t-1} = \lambda\left(d_{i,t}^{*} - d_{i,t-1}\right) + \gamma_t + v_{i,t}$$

$$d_{i,t} = \lambda\alpha + (1-\lambda)d_{i,t-1} + \lambda B' X_{i,t} + \gamma_t + \lambda\eta_i + v_{i,t}$$

$$d_{i,t} = \beta_0 + \beta_1 d_{i,t-1} + B_2' X_{i,t} + \gamma_t + \theta_i + v_{i,t}$$

Empirical Analysis Continue…

Do firms that become ESG rated alter their optimal leverage ratio?

- On average, a firm's optimal market leverage ratio reduces from 15.9% to 12.6% after they become ESG rated
- This reduction is at a magnitude of 20.7% and statistically significant
- Optimal book leverage ratio drops to the level of 21 which is a 12.1 reduction in book leverage when a firm becomes ESG rated
- Hypothesis I is verified

What is the role of the ESG rating signal to the market?

- A signal mechanism to reduce information asymmetry and improve their access to financial markets

Corporate debt structure and leverage ratios differences between ESG and non-ESG rated firms?

- No significant effect of ESG rating on firms' market and book current leverage ratios
- Firms with high level of profits and turnover rate have low leverage ratios
- Firms with an ESG rating tend to redistribute the source of financing from issuing bonds to bank loans
- Hypothesis II verified: ESG rated → disclosure → less information asymmetry

Endogeneity

- Individual firm heterogeneity, omitted variables bias, reverse causality and other

- How do firms' leverage ratio alter when they get their initial ESG ratings?
- Firms tend to restructure their debt immediately after they become rated
- Two-stage least squares estimation
- One standard deviation increase in the natural logarithm of the ESG with controversies score lead to an increase of about 15.9% in bank loans and a decrease of about 12.9% in bond issuing
- GMM estimation: debt redistribution effects of ESG ratings remain valid

Robustness checks

- Replacing the overall ESG combined ratings with the individual ratings from ESG pillars separately
- Issuing bonds is consistently and significantly reduced under each pillar

- Is redistribution effect more pronounced for a certain type of ESG rated firms?
- ESG rated firms with high financial pressure engage mainly in redistributing their borrowing from external to internal sources
- High growth firms limit their use of debt to avoid the underinvestment issues (might arise from high leverage)
- Firms with alternative uses of assets are in need to increase their borrowing
- Larger firms are able to fully exploit the acquired ESG rating and redistribute their debt from bond to bank loans

- One period lagged firm characteristics. The result is statistically significant

- Firms obtain an ESG score above the median are easier to raise funds from banks
- Level of ESG ratings is irrelevant in the decrease of bonds debt

- Full sample. Key redistribution outcome remains valid

Results

- Optimal leverage ratios and information asymmetry are reduced when firms become ESG rated
- ESG rated firms redistributed their financing sources form public debt (bonds issuing) to private debt (bank loans)
- The substitution effect is more pronounced for firms with high financial pressure, low growth opportunities and specialized assets

Reference

Asimakopoulos, P., Asimakopoulos, S., & Li, X. (2023). The role of environmental, social, and governance rating on corporate debt structure. Journal of Corporate Finance, 83, 102488. https://doi.org/10.1016/j.jcorpfin.2023.102488

Thank you for your attention!